

**國立暨南國際大學資訊工程學系雙主修科目一覽表**

**Double-Major Course Catalog of Department of Computer Science and Information Engineering, National Chi Nan University**

本表自(111)學年度申請雙主修起適用 Double-major application (2022 version)

雙主修最低應修學分： 77 學分（系必修49學分+系專業選修28學分） The minimum credits required for a double-major program：77 (including 49 credits in Departmental Required Courses and 28 credits in Elective Courses.)

| 課程代號<br>Course Code | 課程名稱(中文及英文)<br>Course Name(Chinese and English)           | 學分數<br>Credits | 修別<br>Course type<br>(required/elective) | 開課年級<br>Course availability<br>(freshman/sophomore/junior/senior) | 備註<br>Notes |
|---------------------|-----------------------------------------------------------|----------------|------------------------------------------|-------------------------------------------------------------------|-------------|
| 210036              | 計算機概論<br>Introduction to Computer Science                 | 3              | 必                                        | 一上                                                                |             |
| 210003              | 微積分（上）<br>Calculus(1)                                     | 3              | 必                                        | 一上                                                                |             |
| 210013              | 離散數學<br>Discrete Mathematics                              | 3              | 必                                        | 一上                                                                |             |
| 210008              | 微積分（下）<br>Calculus(2)                                     | 3              | 必                                        | 一下                                                                |             |
| 210099              | 程式設計<br>Programming                                       | 3              | 必                                        | 一下                                                                | 擋修PHP網頁程式設計 |
| 210151              | 邏輯設計與實驗（一）<br>Logic Design and Laboratory Experiments (1) | 3              | 必                                        | 一下                                                                |             |
| 210016              | 資料結構與演算法（一）<br>Data Structures and Algorithms ( I )       | 3              | 必                                        | 二上                                                                |             |
| 210152              | 邏輯設計與實驗（二）<br>Logic Design and Laboratory Experiments (2) | 3              | 必                                        | 二上                                                                |             |
| 210022              | 資料結構與演算法（二）<br>Data Structures and Algorithms (II)        | 3              | 必                                        | 二下                                                                |             |
| 210111              | 機率<br>Probability                                         | 3              | 必                                        | 一下                                                                |             |
| 210026              | 線性代數<br>Linear Algebra                                    | 3              | 必                                        | 二下                                                                |             |
| 210025              | 計算機組織與結構<br>Computer Organization and Architecture        | 3              | 必                                        | 三上                                                                |             |
| 210093              | 系統程式<br>System Programming                                | 3              | 必                                        | 三上                                                                |             |
| 210121              | 專題（一）<br>Project Study ( I )                              | 2              | 必                                        | 三上                                                                |             |
| 210122              | 專題（二）<br>Project Study (II)                               | 2              | 必                                        | 三下                                                                |             |
| 210102              | 作業系統<br>Operating Systems                                 | 3              | 必                                        | 三下                                                                |             |
| 210038              | 微算機實驗<br>Microcomputer Systems Lab                        | 3              | 必                                        | 三下                                                                |             |
| 210015              | 數位電子學<br>Digital Electronics                              | 3              | 選                                        |                                                                   |             |
| 130031              | 軟體工程<br>Software Engineering                              | 3              | 選                                        | 3,4,g,p                                                           |             |

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| 210021 | 組合數學<br>Combinatorial Mathematics                        | 3 | 選 | 1,2,3,4 |  |
| 210123 | 自動機與形式語言<br>Automata and Formal Language                 | 3 | 選 | 1,2,3,4 |  |
| 210124 | 數位系統設計<br>Digital System Design                          | 3 | 選 | 1,2,3,4 |  |
| 210125 | 數值方法<br>Numerical Methods                                | 3 | 選 | 1,2,3,4 |  |
| 210048 | 高等演算法（一）<br>Advanced Algorithms (I)                      | 3 | 選 | 3,4,g,p |  |
| 210126 | 電腦視覺與人機介面<br>Computer Vision and Man-machine Interface   | 3 | 選 | 1,2,3,4 |  |
| 210050 | 圖形理論及應用<br>Graph Theory with Applications                | 3 | 選 | 3,4,g,p |  |
| 210052 | 數位影像處理<br>Digital Image Processing                       | 3 | 選 | 3,4,g,p |  |
| 210127 | 資料壓縮<br>Data Compression                                 | 3 | 選 | 3,4,g   |  |
| 210128 | 電腦圖學<br>Computer Graphics                                | 3 | 選 | 3,4,g   |  |
| 210129 | Web技術<br>WEB Technologies                                | 3 | 選 | 1,2,3,4 |  |
| 210060 | 高等演算法（二）<br>Advanced Algorithms (II)                     | 3 | 選 | 3,4,g,p |  |
| 210082 | 虛擬實境<br>Virtual Reality                                  | 3 | 選 | 3,4,g,p |  |
| 210085 | 多媒體通訊<br>Multimedia and Communication                    | 3 | 選 | 3,4,g,p |  |
| 210130 | 具體數學<br>Concrete Mathematics                             | 3 | 選 | 1,2,3,4 |  |
| 210131 | UNIX程式設計<br>UNIX Programming                             | 3 | 選 | 1,2,3,4 |  |
| 210132 | 程式語言<br>Programming Languages                            | 3 | 選 | 1,2,3,4 |  |
| 210096 | 高等計算機結構<br>Advanced Computer Architecture                | 3 | 選 | 3,4,g,p |  |
| 210103 | 生物計算<br>Computational Biology                            | 3 | 選 | 3,4,g,p |  |
| 210133 | 隨機程序<br>Introduction to Fundamental Stochastic Processes | 3 | 選 | 1,2,3,4 |  |
| 210107 | 組合語言<br>Assembly Language                                | 3 | 選 | 3,4,g,p |  |
| 210134 | LISP程式設計<br>LISP Programming                             | 3 | 選 | 1,2,3,4 |  |
| 210135 | UNIX使用入門<br>Introduction to Unix Environment             | 3 | 選 | 1,2,3,4 |  |
| 215021 | 自然語言處理<br>Natural Language Processing (NLP)              | 3 | 選 | 3,4,g,p |  |
| 210136 | 電腦網路通信協定<br>Computer Network Protocols                   | 3 | 選 | 1,2,3,4 |  |
| 215038 | 錯誤更正碼<br>Error Correcting Code                           | 3 | 選 | 3,4,g,p |  |
| 215047 | 資訊檢索<br>Information Retrieval                            | 3 | 選 | 3,4,g,p |  |
| 215049 | 史料與史學<br>Historical Sources and Historiography           | 3 | 選 | 3,4,g,p |  |
| 219066 | 嵌入式系統實作<br>Embedded System Implementation                | 3 | 選 | 3,4,g,p |  |

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| 219068 | 非擬真成像技術<br>Non Photo-realistic Rendering                                                          | 3 | 選 | 3,4,g,p | 擋修電腦圖學及高等電腦圖學 |
| 210137 | 科技英文(一)<br>Technical English(I)                                                                   | 3 | 選 | 1,2,3,4 |               |
| 219091 | 智慧型演化計算<br>Intelligent Evolutionary Computation                                                   | 3 | 選 | 3,4,g,p |               |
| 210140 | 智慧型行動裝置軟體設計<br>Intelligent Mobile Device Programming                                              | 3 | 選 | 1,2,3,4 |               |
| 219093 | 智慧型系統實務(一)<br>Intelligent System Practice (1)                                                     | 2 | 選 | 3,4,g,p |               |
| 219096 | Linux系統管理<br>Linux System Administration                                                          | 3 | 選 | 3,4,g,p |               |
| 219097 | 進階Linux系統管理<br>Advanced Linux System Administration                                               | 3 | 選 | 3,4,g,p |               |
| 219101 | 語言學習與科技<br>Language Learning and Technology                                                       | 3 | 選 | 3,4,g,p |               |
| 219103 | 機器學習<br>Machine Learning                                                                          | 3 | 選 | 3,4,g,p |               |
| 219106 | 適地性文創應用與服務系統<br>Location-based Cultural & Creative Application and Service Systems                | 3 | 選 | 3,4,g,p |               |
| 219107 | 智慧型文化導向行動導覽服務創意平台技術<br>Intelligent Web Mobile Platform for Culture-Oriented and Creative Services | 3 | 選 | 3,4,g,p |               |
| 219108 | 行動式3D電腦繪圖<br>Mobile 3D Computer Graphics                                                          | 3 | 選 | 3,4,g,p | 擋修電腦圖學及高等電腦圖學 |
| 219110 | 地方文獻編纂與習作<br>Local Bibliography Compilation and Practice                                          | 3 | 選 | 3,4,g,p |               |
| 210141 | 3D建模與互動多媒體應用<br>3D Modeling and Interactive Multimedia Applications                               | 3 | 選 | 1,2,3,4 |               |
| 219113 | 網際網路電話加值服務<br>IP Telephony Service Creation Technology                                            | 3 | 選 | 3,4,g,p |               |
| 210142 | 數值分析<br>Numerical Analysis                                                                        | 3 | 選 | 1,2,3,4 |               |
| 210143 | Web服務和應用<br>Web Services and Applications                                                         | 3 | 選 | 1,2,3,4 |               |
| 210144 | 通訊導論<br>Introduction to Communications                                                            | 3 | 選 | 3,4,g,p |               |
| 210119 | 網際網路程式設計<br>WWW Programming Design                                                                | 3 | 選 | 3,4,g,p |               |
| 219076 | 智慧型系統實務（三）<br>Intelligent System Practice (3)                                                     | 2 | 選 | 3,4,g,p |               |
| 219099 | 雲端系統設計<br>Design of Cloud Computing System                                                        | 3 | 選 | 3,4,g,p |               |
| 219115 | 音樂,大腦,與科技<br>Music,Brain,Science and Technology                                                   | 3 | 選 | 3,4,g,p |               |
| 219116 | 英文寫作, 口說, 簡報 (一)<br>On English Writing, Speaking andPresentation (1)                              | 3 | 選 | 3,4,g,p |               |
| 219117 | 英文寫作, 口說, 簡報 (二)<br>On English Writing, Speaking andPresentation (2)                              | 3 | 選 | 3,4,g,p |               |
| 210146 | 基礎程式設計<br>Computer Fundamentals & Programming                                                     | 1 | 選 | 1,2,3,4 |               |

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| 219121 | 尖端資訊科技國際課程<br>International Studies on Leading<br>Information Technologies                                                      | 1 | 選 | 3,4,g,p |               |
| 219118 | 史學研究與資訊科技跨領域合作實務<br>When History Meets Computer Science:<br>Theory and Practice                                                 | 3 | 選 | 3,4,g,p |               |
| 219130 | 安卓程式設計<br>Android Programming                                                                                                   | 3 | 選 | 3,4,g,p |               |
| 219122 | QNAP應用程式設計<br>Linux Programming for QNAP                                                                                        | 3 | 選 | 3,4,g,p |               |
| 219123 | 開源電子原型平台軟硬體整合設計與<br>實作<br>software / hardware co-design &<br>implementation for open-source electronics<br>prototyping platform | 3 | 選 | 3,4,g,p |               |
| 210147 | 基礎程式設計 (二)<br>C++ Programming Fundamentals                                                                                      | 1 | 選 | 1,2,3,4 |               |
| 219128 | 數位音樂學習與創作<br>Digital Music Learning and Creation                                                                                | 3 | 選 | 3,4,g,p |               |
| 219133 | 國際IT科技發展趨勢<br>Global Trend of Information Technology<br>Development                                                             | 2 | 選 | 3,4,g,p |               |
| 219134 | 尖端資訊科技國際課程 (二)<br>International Studies on Leading<br>Information Technologies(2)                                               | 2 | 選 | 3,4,g,p |               |
| 210148 | 專業實習<br>Specialized Internship                                                                                                  | 3 | 選 | 4       |               |
| 210149 | 企業實習<br>Enterprise Internship                                                                                                   | 6 | 選 | 4       |               |
| 219135 | 交響樂團與數位應用<br>Orchestra and Its Digital Applications                                                                             | 2 | 選 | 3,4,g,p |               |
| 219136 | 音樂科技專題策展<br>Capstone Project on Music Technology                                                                                | 2 | 選 | 3,4,g,p |               |
| 219137 | 數位錄音錄影工程<br>Digital Audio/Video Recording and<br>Engineering                                                                    | 2 | 選 | 3,4,g,p |               |
| 219138 | 音樂, 大腦, 與科技<br>Music, Brain, Science and Technology                                                                             | 2 | 選 | 3,4,g,p |               |
| 219139 | 數位音樂學習與創作<br>Digital Music Learning and Creation                                                                                | 2 | 選 | 3,4,g,p |               |
| 219140 | PHP 網頁程式設計<br>PHP Web Programming                                                                                               | 3 | 選 | 3,4,g,p | 需先修210099程式設計 |
| 219141 | 計算智能<br>Computational Intelligence                                                                                              | 3 | 選 | 3,4,g,p |               |
| 219142 | 大數據系統實務<br>Big Data System and Practice                                                                                         | 3 | 選 | 3,4,g,p |               |
| 219143 | 深度學習<br>Deep Learning                                                                                                           | 3 | 選 | 3,4,g,p |               |
| 210009 | 電子電路<br>Electrical Circuits                                                                                                     | 3 | 選 | 1,2,3,4 |               |
| 210086 | 工程數學<br>Engineering Mathematics                                                                                                 | 3 | 選 | 1,2,3,4 |               |
| 210115 | 邏輯設計<br>Logic Design                                                                                                            | 3 | 選 | 1,2,3,4 |               |
| 210040 | 編譯器<br>Compilers                                                                                                                | 3 | 選 | 3,4,g,p |               |

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| 210076 | 資料庫系統<br>Database System                                                                                       | 3 | 選 | 3,4,g,p |                  |
| 210145 | 計算機網路<br>Computer Networks                                                                                     | 3 | 選 | 1,2,3,4 |                  |
| 210023 | 數位電路實驗<br>Digital Circuits Lab                                                                                 | 1 | 選 | 1,2,3,4 |                  |
| 210029 | 微計算機系統<br>Microcomputer Systems                                                                                | 3 | 選 | 1,2,3,4 |                  |
| 210150 | 基本電學<br>Basic Electricity                                                                                      | 3 | 選 | 1,2,3,4 |                  |
| 210153 | MFC視窗程式設計<br>Windows MFC Programming                                                                           | 3 | 選 | 1,2,3,4 | 需先修程式設計          |
| 210154 | 個人電腦概論<br>Information Technology Essentials                                                                    | 3 | 選 | 1,2,3,4 | 需先修計算機概論         |
| 219146 | 程式練習及訓練與競賽課程<br>Programming Practice, Training, and Contest Course                                             | 3 | 選 | 3,4,g,p |                  |
| 210155 | Web程式設計<br>Web Programming                                                                                     | 3 | 選 | 3,4,g,p |                  |
| 210156 | JavaScript程式設計<br>JavaScript Programming                                                                       | 3 | 選 | 1,2,3,4 | 需先修程式設計          |
| 219148 | SQL程式設計<br>SQL Programming                                                                                     | 3 | 選 | 3,4,g,p | 大學部必須先修通過資料庫系統   |
| 219149 | 程式練習及訓練與競賽（二）<br>Programming Practice, Training, and Contest Course (II)                                       | 3 | 選 | 3,4,g,p |                  |
| 219150 | 前端程式設計<br>Frontend Programming                                                                                 | 3 | 選 | 3,4,g,p | 大學部必須先修通過Web程式設計 |
| 219151 | IT科技日文<br>IT/Technical Japanese                                                                                | 3 | 選 | 3,4,g,p | 需先修日文一(上)、日文一(下) |
| 219152 | Python網頁擷取程式設計<br>Web Scraping with Python                                                                     | 3 | 選 | 3,4,g,p | 需先修計算機概論         |
| 215050 | MXNET之深度學習實驗<br>MXNET-based Deep Learning Lab                                                                  | 3 | 選 | 2,3,4,g |                  |
| 219155 | Python機器學習實務<br>Machine Learning Practice with Python                                                          | 3 | 選 | 3,4,g,p |                  |
| 219156 | iOS App 程式開發<br>iOS App Programming                                                                            | 3 | 選 | 3,4,g,p |                  |
| 215051 | C 語言教學探討<br>Discussion and Inquiry of Learning C                                                               | 3 | 選 | 3,4,g   |                  |
| 219157 | FPGA實作深度學習方法<br>FPGA Implementation of Deep Learning Methods                                                   | 3 | 選 | 3,4,g,p | 需先修邏輯設計或邏輯設計與實驗  |
| 215052 | VR/AR應用實作<br>VR/AR - Implementation and Applications                                                           | 3 | 選 | 3,4,g,p | 需先修電腦圖學          |
| 219159 | 情境模擬控制系統之軟硬體整合應用<br>實務<br>Hardware / Software Integration & Practice for Situational Simulation Control System | 3 | 選 | 3,4,g,p |                  |
| 219160 | 智慧型積木機器人程式設計實務 EV3<br>Intelligent Lego Robot Programming                                                       | 3 | 選 | 3,4,g,p |                  |
| 219161 | 音樂程式設計<br>Music Programming                                                                                    | 3 | 選 | 3,4,g,p |                  |
| 210106 | 電腦與資訊安全<br>Computer and Information Security                                                                   | 3 | 選 | 3,4,g,p |                  |
| 210157 | 企業實習240(一)<br>Enterprise Internship 240(一)                                                                     | 3 | 選 | 4       |                  |
| 210158 | 企業實習240(二)<br>Enterprise Internship 240(二)                                                                     | 3 | 選 | 4       |                  |

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| 210159                               | 企業實習(二)<br>Corporate Internship (2)                           | 6 | 選 | 4         |  |
| 219169                               | 資訊科學特論<br>Special Topics in Computer Science                  | 3 | 選 | 3,4,g,p   |  |
| 219168                               | 物聯網晶片開發實務<br>Practices of Internet of Things Chip Development | 3 | 選 | 3,4,g,p   |  |
| 215058                               | 互動音樂程式設計與創作<br>Interactive Music Program and Creation         | 1 | 選 | 1,2,3,4,g |  |
| 210161                               | Linux系統管理<br>Linux System Administratio                       | 3 | 選 | 1,2,3,4   |  |
| 219171                               | 深度學習實驗<br>Deep Learning Lab                                   | 3 | 選 | 3,4,g,p   |  |
| 215056                               | 程式設計與運算思維<br>Programming and Computational Thinking           | 3 | 選 | 1,2,3,4,g |  |
| 215057                               | 程式設計與美學創作<br>Creative Coding and Aesthetic                    | 3 | 選 | 1,2,3,4,g |  |
| 專業選修28學分：開課年級依各課程決定，修課建議請參考專業選修課程地圖。 |                                                               |   |   |           |  |